

Question	Marking details	Marks available					
		AO1	AO2	AO3	Total	Maths	Prac
7	Indicative content Anaesthetic A A1 Anaesthetic A {blocks Na⁺ channels / Na⁺ channels don't open} A2 Threshold (potential) is not met/ ref to all or nothing A3 (membrane) not depolarised A4 Action potential is not generated A5 So impulse does not reach {pain centre/ brain} Anaesthetic B B1 Anaesthetic B Blocks Ca²⁺ channels/ Ca²⁺ channels do not open B2 (Ca²⁺ can't enter) so synaptic vesicles don't fuse with presynaptic membrane B3 Neurotransmitter not released into synaptic cleft/ no exocytosis B4 Postsynaptic membrane not depolarised B5 No action potential generated B6 So impulse does not reach {pain centre/ brain} Compare with reflex arc C1 Both have sensory neurones/ both have three neurones C2 (With cell bodies) in ganglia C3 Outside CNS C4 Pain pathway has no motor neurones/ shorter motor neurone/ motor neurone in brain Accept reverse answer for reflex C5 Pain pathway has no relay neurones/ relay neurone in brain/longer relay neurone Accept reverse answer for reflex C6 Pain pathway has no effector/ effector is located in the brain C7 Involves brain not spinal cord	5	4		9		